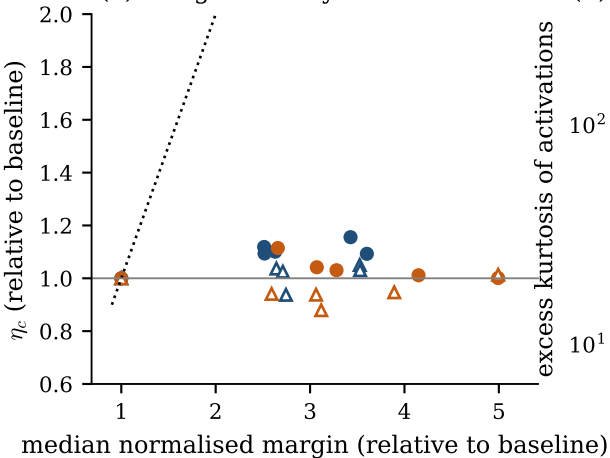
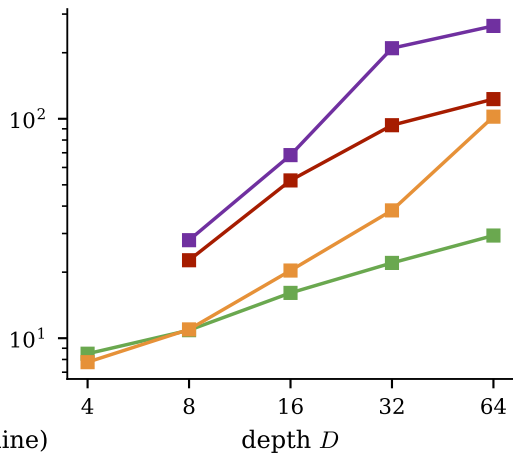


(a) margins barely move the floor



(b) activation tails grow with depth (CNNs)



● PTQ, depth 8 ▲ QAT, depth 8 $\eta_c \propto \text{margin}$
 ● PTQ, depth 16 ▲ QAT, depth 16

■ CNN res. $s = 1/\sqrt{D}$ ■ CNN res. $s = 1$
 ■ CNN res. $s = 1/\sqrt{8}$ ■ CNN plain (Dirac)